

12345678

RevNoRevision noteDateSignatureChecked

TODO

- * 8xI2C mux INT signals on a header (GPIO expander)
- * 2xI2C mux SDA/SDC signals on a header
- * CAN termination jumpers.
- * Check courtyards of components.
- * Diode footprints/model
- * upgrade OT1-6 Dual MOSFETs

MAYBE

- * connector for front-panel USB socket - should probably go to raspberry-pi - need a small break-out PCB in that case.
- * reverse polarity protection, need multiple high-current fets like SUD50P06-15-GE3.
- * VUSB sense input.

DONE

- * Load-cell amplifier (HX717)
- * Additional 5V rail for onboard and up-lights.
- * Raspberry Pi 5 power connector
- * Supercap/RTC osc - deleted.
- * Models for JST-XH (e.g 'B4B-XH-A(LF)(SN)')
- * Ethernet PHY
- * MOSFET + connector for external ethernet link leds.
- * 3V3 power from core board
- * LEDs for power rails.
- * RGB LED output, for up-lights
- * 2x vacuum sensors.
- * Head RS422 (non-isolated power).
- * Analog mux (2x1 in 4 out).

ItemrefQuantityTitle/Name, designation, material, dimension etcArticle No./Reference

Designed byChecked byApproved by - date- 00/00/00File nameDate00/00/00Scale1:1

Designed byDominic Clifton

Edition0Sheet1/1

A

B

C

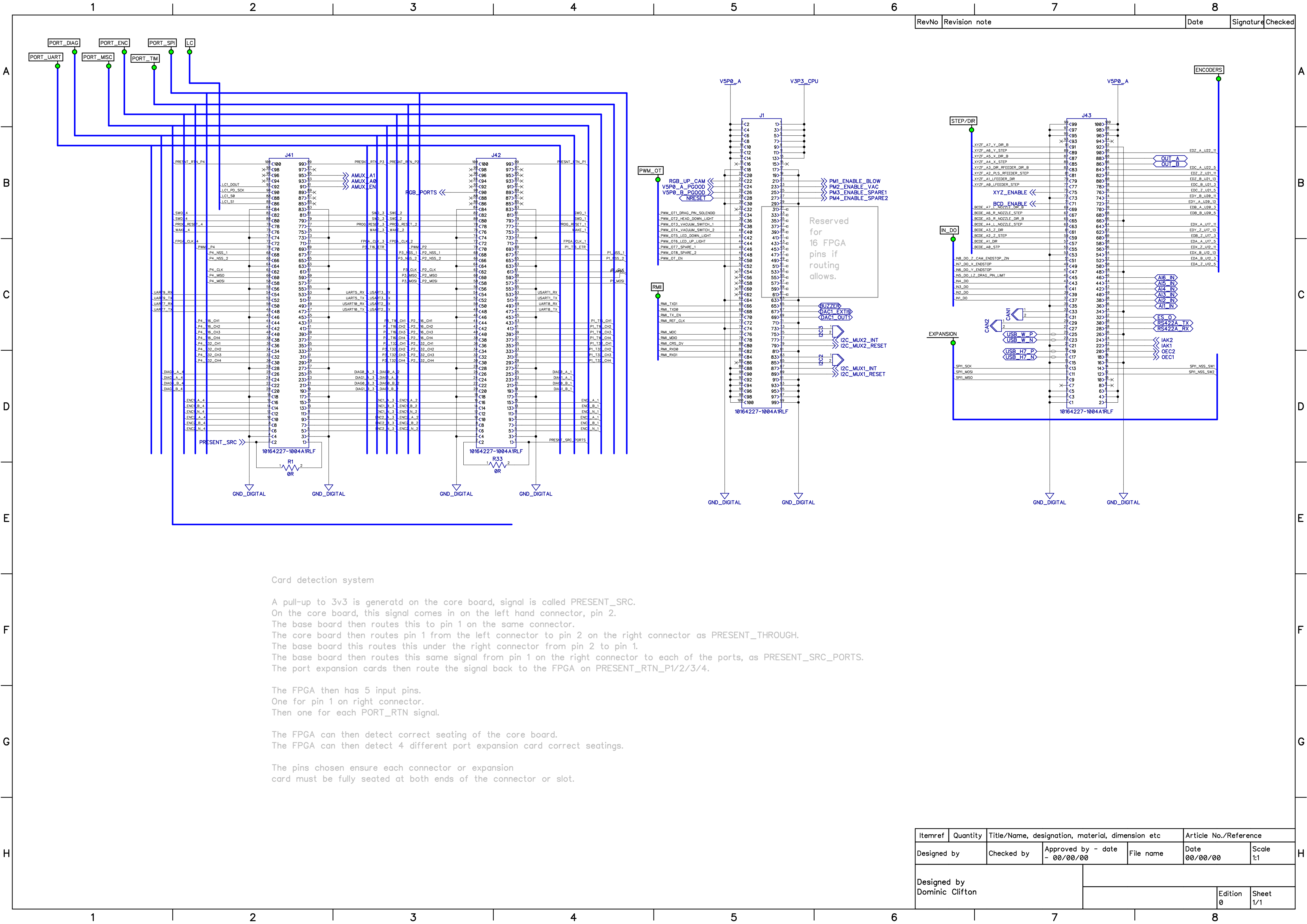
D

E

F

G

H



Card detection system

A pull-up to 3v3 is generatd on the core board, signal is called PRESENT_SRC. On the core board, this signal comes in on the left hand connector, pin 2. The base board then routes this to pin 1 on the same connector. The core board then routes pin 1 from the left connector to pin 2 on the right connector as PRESENT_THROUGH. The base board then routes this under the right connector from pin 2 to pin 1. The base board then routes this same signal from pin 1 on the right connector to each of the ports, as PRESENT_SRC_PORTS. The port expansion cards then route the signal back to the FPGA on PRESENT_RTN_P1/2/3/4.

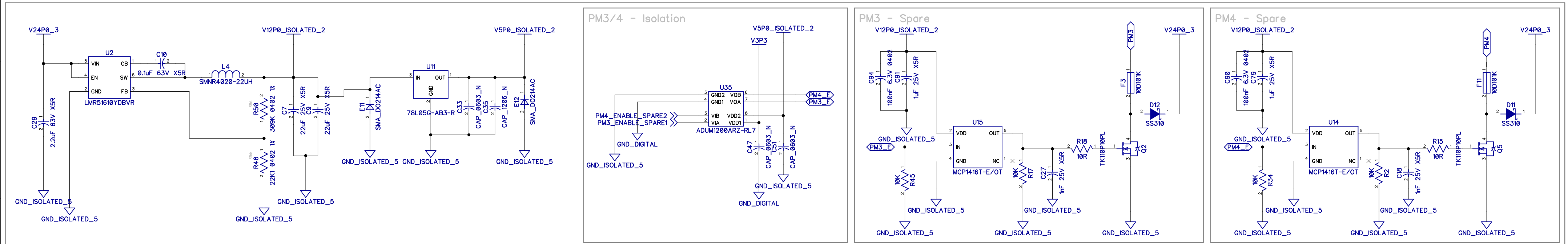
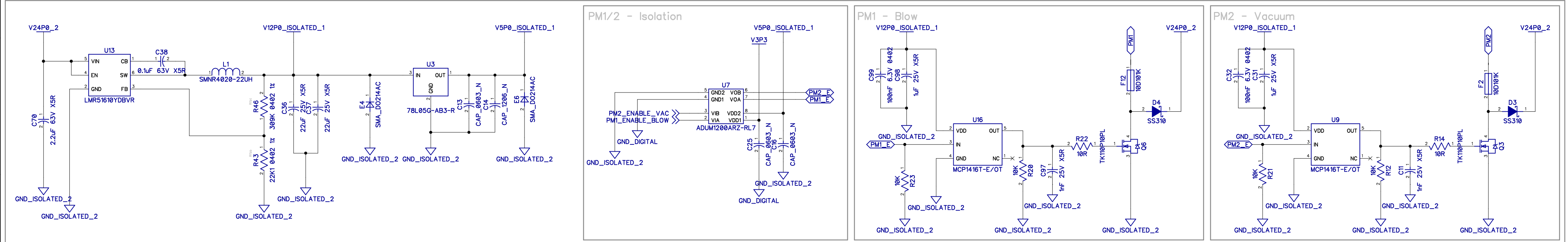
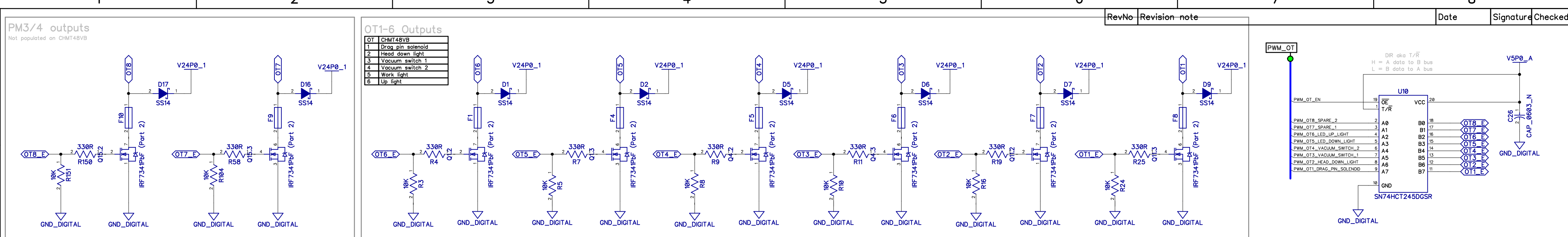
The FPGA then has 5 input pins. One for pin 1 on right connector. Then one for each PORT_RTN signal.

The FPGA can then detect correct seating of the core board. The FPGA can then detect 4 different port expansion card correct seatings.

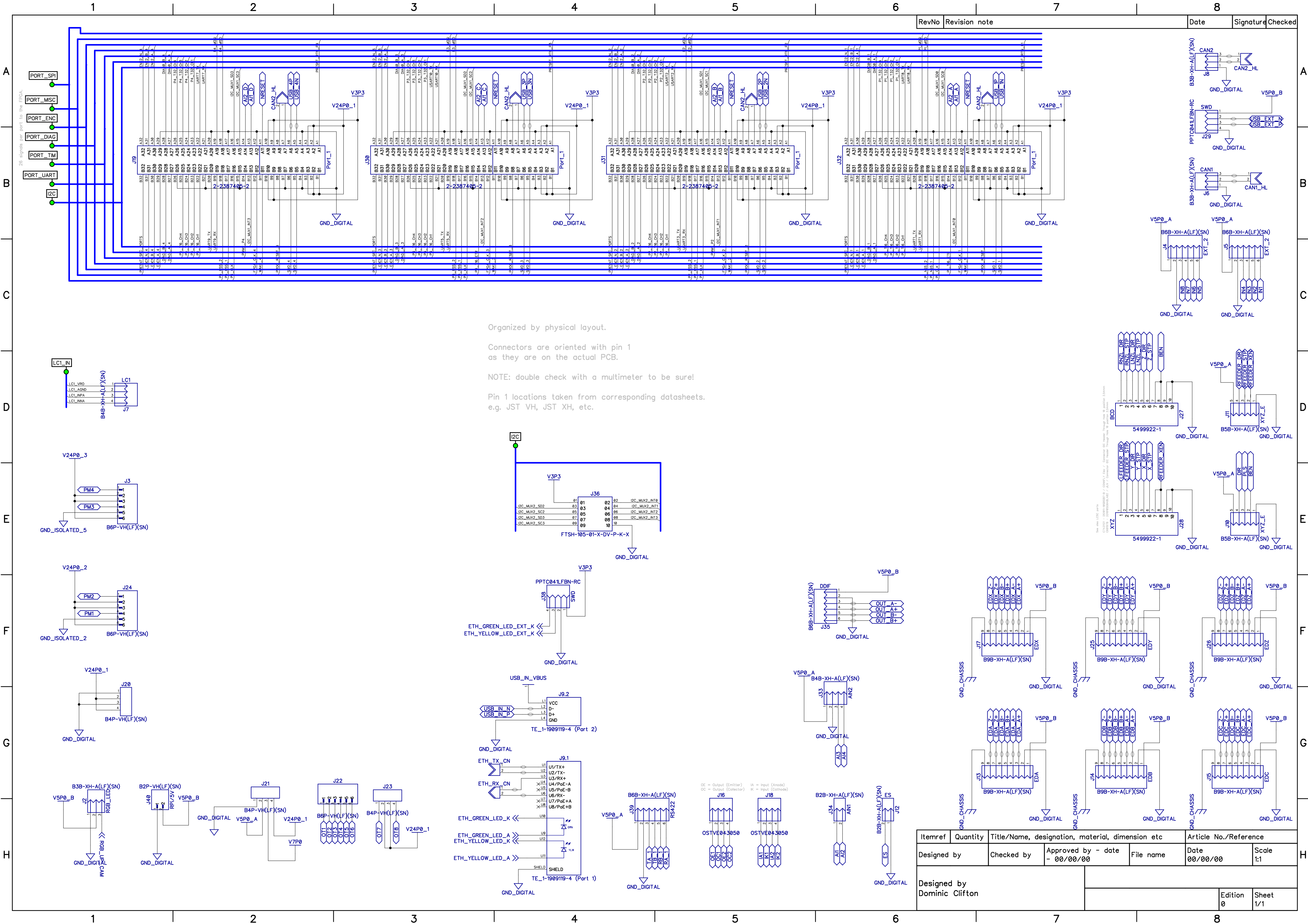
The pins chosen ensure each connector or expansion card must be fully seated at both ends of the connector or slot.

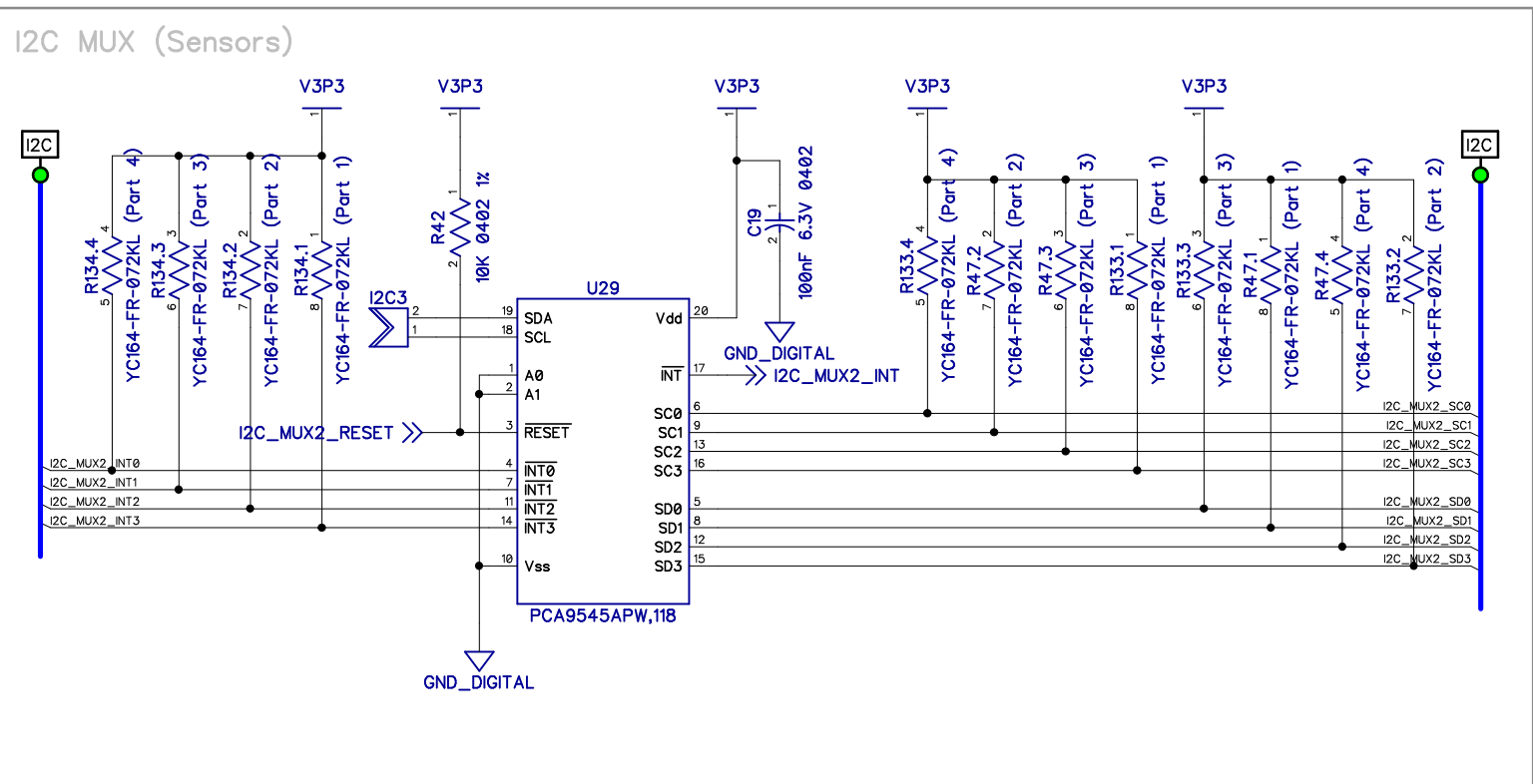
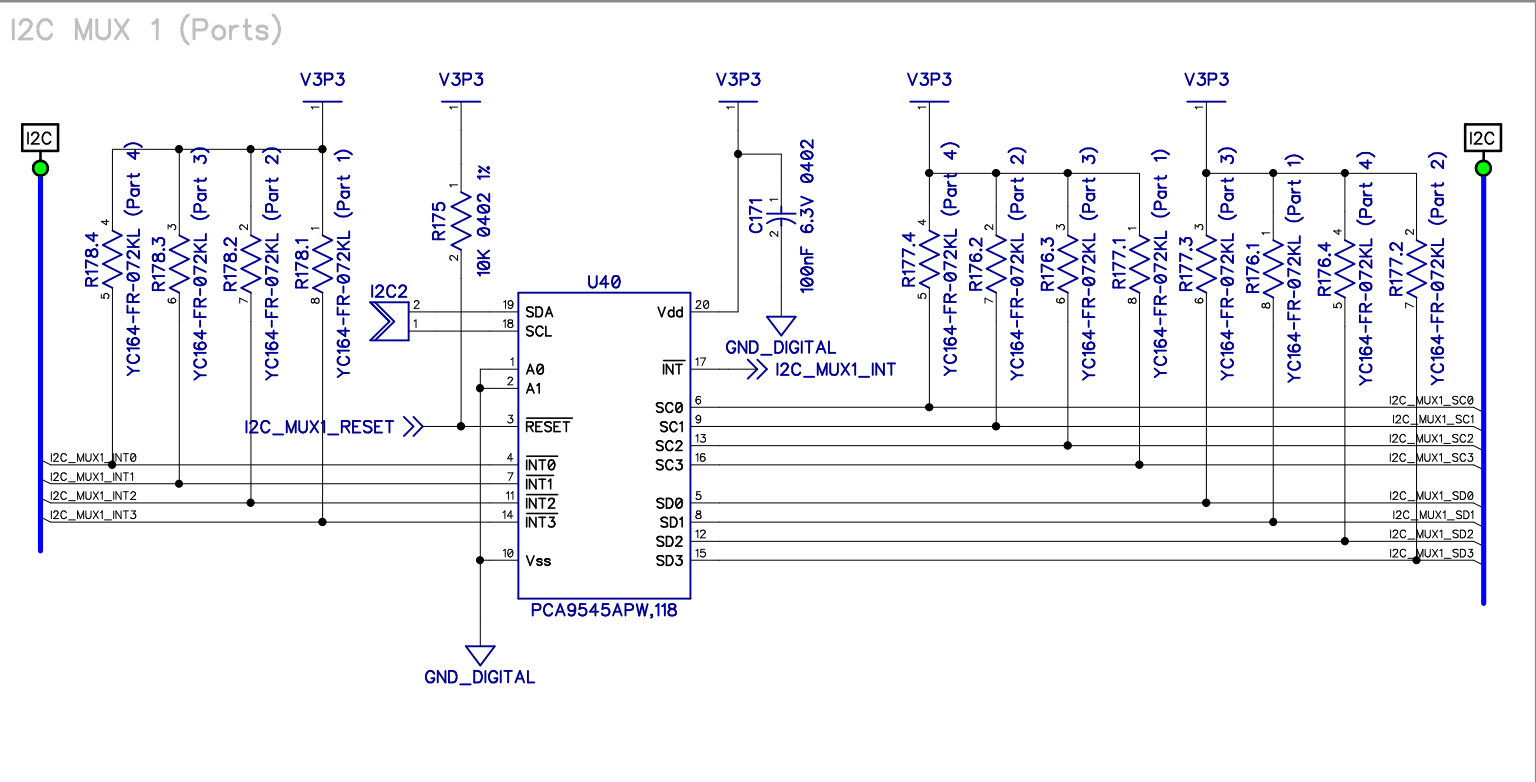
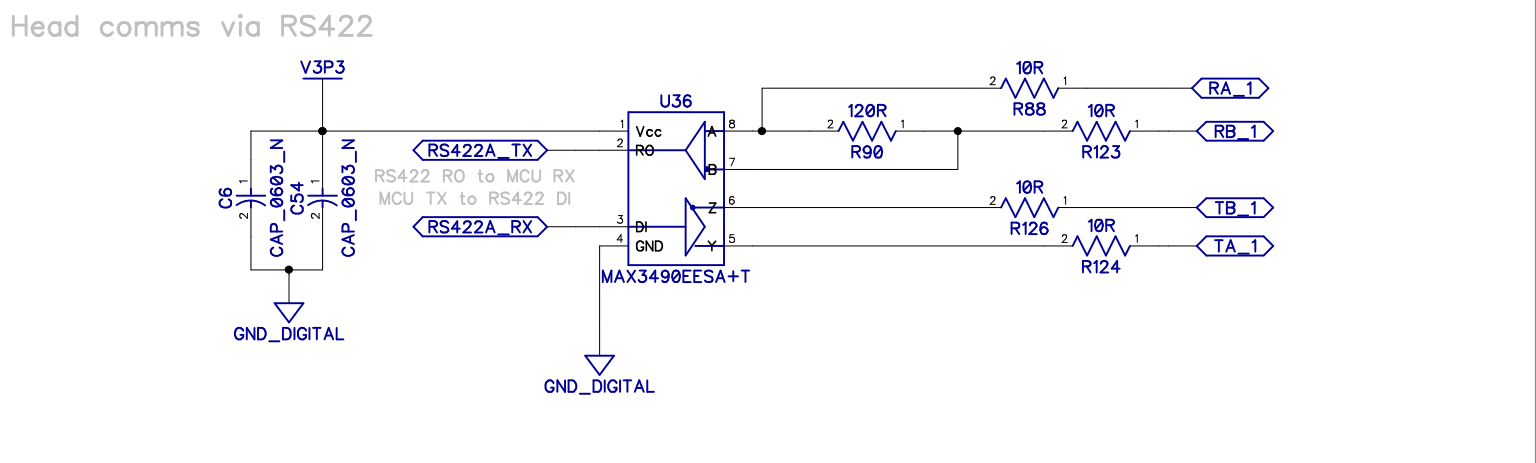
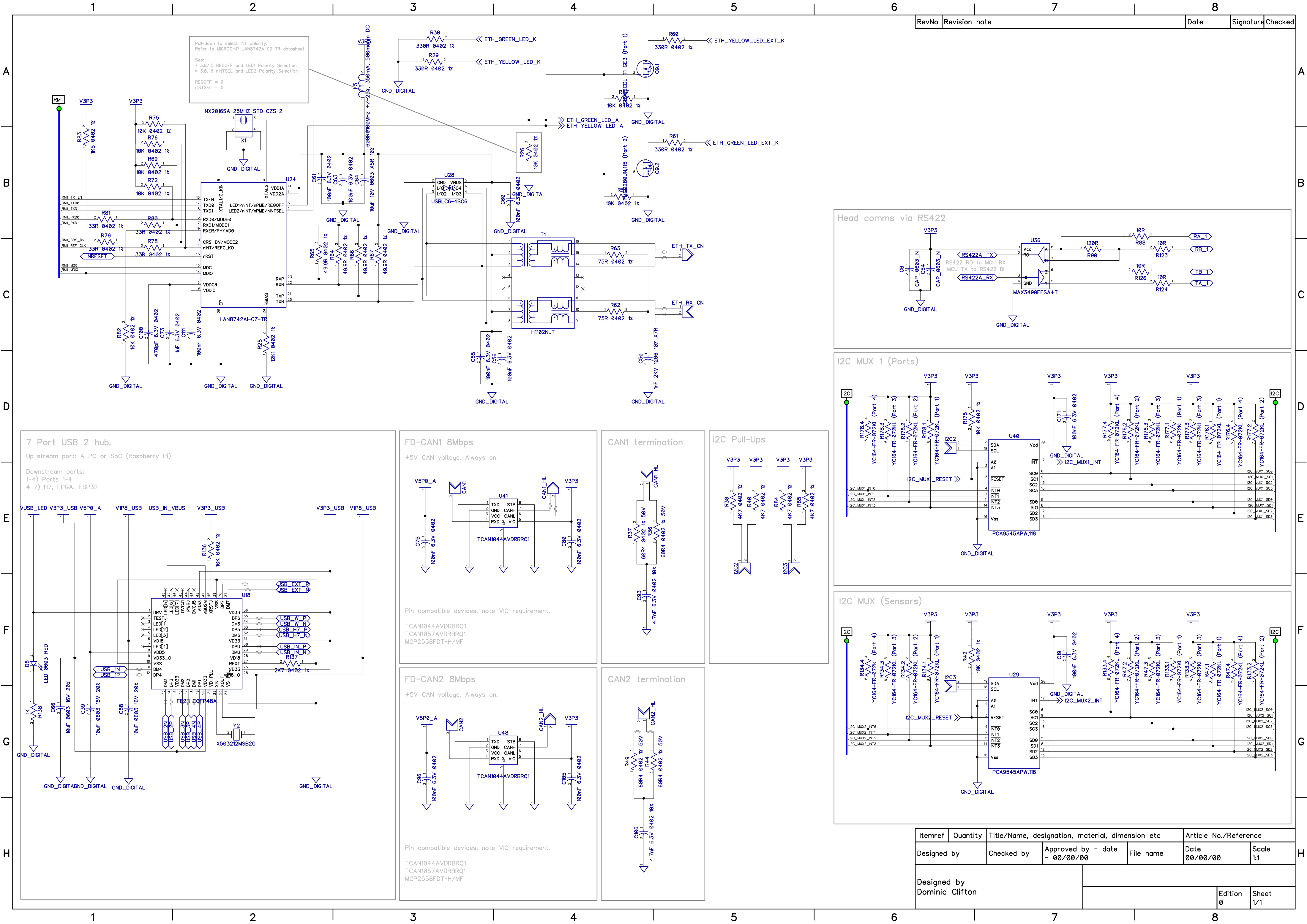
RevNo	Revision note	Date	Signature	Checked
-------	---------------	------	-----------	---------

Itemref	Quantity	Title/Name, designation, material, dimension etc			Article No./Reference	
Designed by	Checked by	Approved by - date	File name	Date	Scale	
Designed by Dominic Clifton		00/00/00		00/00/00	1:1	
				Edition	Sheet	
				0	1/1	



Itemref	Quantity	Title/Name, designation, material, dimension etc			Article No./Reference	
Designed by		Checked by	Approved by - date - 00/00/00	File name	Date 00/00/00	Scale 1:1
Designed by Dominic Clifton						
					Edition 0	Sheet 1/1





Itemref	Quantity	Title/Name, designation, material, dimension etc			Article No./Reference	
Designed by	Checked by	Approved by - date	File name	Date	Scale	
Designed by Dominic Clifton		00/00/00		00/00/00	1:1	
				Edition	Sheet	
				0	1/1	

